Notes: Müller & Verner (RES, 2024)

Credit Allocation and Macroeconomic Fluctuations

Contents

1 Big picture

- **Question.** Why do some credit booms end in financial crises and growth slowdowns, while others do not? The paper's answer is that *who receives the credit* matters as much as how much credit grows.
- **Setting.** A new long-run cross-country database on sectoral private credit covering 117 countries from 1940 to 2014, with the main macro panel restricted to 75 countries with population above one million in 2000.
- **Core empirical object.** The paper splits private credit into household credit and five broad firm sectors, then aggregates firms into *tradable* sectors (agriculture; manufacturing and mining) and *non-tradable* sectors (construction and real estate; trade, accommodation, and food; transport and communication).
- **Main descriptive fact.** During major credit booms, credit systematically reallocates toward non-tradable firms and households. These sectors account for about 70% of total lending growth during booms, and the non-tradable plus household credit share rises in four out of five booms.
- **Main predictive result.** Credit expansion toward non-tradable firms predicts medium-run GDP slowdowns and a higher probability of systemic banking crises. Credit expansion toward tradable firms does *not*; if anything, it predicts stronger future productivity and output growth.
- **Boom type distinction.** Five years after the start of a credit boom biased toward non-tradables or households, real GDP is about 5% lower than after a tradable-biased credit boom.
- **Mechanism.** Non-tradable firms appear more financially fragile: they are smaller, rely more on real-estate-backed lending, are more exposed to domestic-demand and collateral feedback loops, and default more during crises.
- **What the paper is really saying.** "Bad" booms are not just large credit booms. They are booms in which credit flows toward sectors that are financially constrained, collateral sensitive, and relatively low productivity. "Good" booms are more associated with lending to tradables, especially manufacturing-like sectors.
- **Contribution.** The paper brings sectoral heterogeneity into the macro-credit literature. It helps reconcile why the literature sometimes finds that firm credit is benign and sometimes finds that credit booms end badly: it depends on *which* firms are borrowing.
- **Important limitation.** The paper is explicit that its mechanism evidence is not fully causal. The contribution is a large, disciplined reduced-form macro panel plus theory-consistent mechanism tests, not a clean instrument for sectoral credit shocks.

2 Data and measurement (key objects)

2.1 The new sectoral credit database

- **Coverage.** The database covers private credit by sector for 117 countries from 1940 to 2014. In total, it contains 5,436 country-year observations with sectoral breakdowns and about 89,019 country-sector-year observations.

- **Comparison to existing sources.** Relative to Jorda et al. (2016a), IMF GDD, and BIS sectoral series, the database is both longer and much more disaggregated. The average country-year contains 16 sector observations, versus 2 or 3 in earlier datasets.
- **Total credit extension.** The authors also extend total private credit series to 189 countries, with some series starting as early as 1910.
- **Sources.** The data are assembled from more than 600 sources, especially central-bank and statistical-office publications, plus BIS, IMF IFS, IMF GDD, and digitized historical IMF IFS volumes.

2.2 Measurement concept and harmonization

- **What is measured.** End-of-period outstanding claims of financial institutions on the domestic private sector. In most countries, this is mainly bank and housing-finance loans, including foreign-currency loans where relevant.
- **Debt instruments.** The data sometimes include bond exposures held on financial institutions' balance sheets, but in practice domestic private credit is overwhelmingly loan-based.
- **Institutional coverage.** The authors try to cover the whole financial system, but the data are predominantly lending by deposit-taking institutions such as commercial banks, savings banks, and housing-finance institutions.
- **Sector classification.** They follow the System of National Accounts to separate households from corporations and use ISIC Revision 4 to map industries.
- **Break adjustment.** Because long-run credit series often shift due to reporting changes, the paper codes methodological breaks country by country and chain-links series to preserve internal consistency.

2.3 Main sector aggregates used in the paper

- **Households.** Household credit is always reported separately when available.
- **Firm sectors.** The main firm groups are agriculture; manufacturing and mining; construction and real estate; wholesale and retail trade, accommodation, and food; and transport and communication.
- **Tradables vs. non-tradables.** Tradables are agriculture plus manufacturing/mining. Non-tradables are construction and real estate; trade, accommodation, and food; and transport and communication.
- **Main estimation sample.** 75 countries with population above one million in 2000; variables are winsorized at the 1st and 99th percentiles.

2.4 Auxiliary data used for mechanisms and controls

- **Firm size and collateral.** Small-firm shares come from OECD Structural Business Statistics. Mortgage or real-estate-collateral shares are assembled for Denmark, Latvia, Switzerland, Taiwan, and the U.S.

- **Macro variables.** GDP, prices, exchange rates, labour productivity, total factor productivity, and sectoral value added come from World Bank, Penn World Tables, IMF, GGDC, EU KLEMS, WIOD, OECD STAN, and related national sources.
- **Crisis data.** Systemic banking-crisis starts come mainly from Baron, Verner, and Xiong (2021), supplemented with Laeven and Valencia (2018).
- **House prices and bond issuance.** House-price data come from BIS, OECD, and Dallas Fed sources; bond issuance comes from SDC Platinum.

2.5 Two key cross-sector facts that matter for the mechanism

- **Non-tradable firms are smaller.** The small-firm share is 0.90 in non-tradables versus 0.79 in tradables.
- **Non-tradable firms rely more on real-estate collateral.** The mortgage-backed or real-estate-secured share is 0.36 in non-tradables versus 0.19 in tradables.
- **Non-tradables are lower productivity-growth sectors.** Average labour-productivity growth is 2.65% in non-tradables versus 5.03% in tradables; average TFP growth is 0.51% versus 2.02%.
- **Interpretation.** These facts motivate the idea that non-tradable firms are more financing constrained during booms and more fragile during busts.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Credit-boom classification and boom composition

Although not written as a numbered equation in the paper, the key event definition is:

$$\text{Boom}_{it} = \mathbf{1}\{\widetilde{d}_{it} > \sigma_i(\widetilde{d}_{it})\}, \tag{1}$$

where $\widetilde{d}_{it}$ is total private credit-to-GDP detrended with the Hamilton (2018) filter using a four-year horizon.

The tradable-credit share used to classify boom types is

$$s_{it}^{\mathrm{T}} = \frac{d_{it}^{\mathrm{T}}}{d_{it}^{\mathrm{T}} + d_{it}^{\mathrm{NT}} + d_{it}^{\mathrm{HH}}}. \tag{2}$$

- **Interpretation.** A major credit boom is a period in which credit-to-GDP is unusually high relative to its recent trend. The paper finds 113 such boom episodes.
- **Boom types.** A boom is *tradable-biased* if the tradable-credit share rises over the previous five years, and *non-tradable-biased* otherwise.
- **Why this matters.** The paper's central claim is not just that aggregate credit matters, but that the allocation of the boom across sectors predicts whether it ends badly.

3.2 Credit reallocation during booms

The main sector-level regression for credit growth during booms is

$$\Delta_3 \ln(d_{i,s,t}) = \alpha_i + \beta_1 \text{Boom}_{i,t} + \beta_2 \big(\text{Boom}_{i,t} \times \text{High Characteristics}_s\big) + \varepsilon_{i,s,t}. \tag{3}$$

- **Dependent variable.** Three-year growth in sectoral credit-to-GDP.
- **Regressor of interest.** The interaction term checks whether booms disproportionately raise credit in sectors that are more non-tradable, have more small firms, or rely more on real-estate collateral.
- **Expected sign.** If easy credit flows toward constrained and collateral-sensitive sectors, then $\beta_2 > 0$.
- **Result.** The interaction coefficients are strongly positive: about 7.1 for non-tradables and small-firm sectors, and about 5.1 for sectors with high mortgage shares.
- **Meaning.** Booms are not broad-based leverage increases. They reallocate credit toward precisely the sectors theory says should respond most to easier financing conditions.

3.3 Event-study growth around different boom types

To compare tradable-biased and non-tradable-biased booms, the paper estimates

$$y_{i,t+h} - y_{i,t-1} = \alpha_i^h + \beta_{\text{T}}^h \text{Boom}_{i,t}^{\text{T}} + \beta_{\text{NT}}^h \text{Boom}_{i,t}^{\text{NT}} + \varepsilon_{i,t+h}, \qquad h = -5, \dots, 5. \tag{4}$$

- **Dependent variable.** Cumulative real GDP growth around the credit-boom event.
- **Interpretation.** The coefficients trace the boom-bust path of output for the two types of booms relative to normal times.
- **Result.** Both boom types are expansionary initially, but only non-tradable-biased booms are followed by a sharp reversal. Five years after boom start, real GDP is about 5% lower than after a tradable-biased boom.
- **Economic meaning.** This is the cleanest reduced-form version of the paper's "good boom / bad boom" distinction.

3.4 Local projections for future GDP after sectoral credit innovations

The baseline dynamic specification is a local projection:

$$\Delta_h y_{i,t+h} = \alpha_i^h + \sum_{k \in K} \sum_{j=0}^{J} \beta_{h,j}^k \Delta d_{i,t-j}^k + \sum_{j=0}^{J} \gamma_{h,j} \Delta y_{i,t-j} + \varepsilon_{i,t+h}, \qquad h = 1, \dots, H. \tag{5}$$

- **Dependent variable.** Real GDP growth from year t to $t + h$.
- **Core regressors.** One-year changes in credit-to-GDP for different sectors.
- **Implementation.** The paper uses $J = 5$ lags and a horizon of $H = 10$, with Driscoll-Kraay standard errors and also reports two-way clustering by country and year.

- **Interpretation.** The sequence $\{\beta_{h,0}^{k}\}$ is the impulse-response-style predictive path of GDP after a credit expansion in sector k.
- **Main result.** A one-percentage-point innovation in non-tradable credit-to-GDP predicts about 0.8% lower cumulative GDP over the next five years; a similar tradable-sector innovation predicts about 0.6% higher GDP over five years and 2.1% over ten years.

3.5 Medium-run panel regressions for GDP growth

The paper also estimates a simpler medium-run regression:

$$\Delta_3 y_{i,t+h} = \alpha_i^h + \beta_h^{\mathrm{NT}} \Delta_3 d_{it}^{\mathrm{NT}} + \beta_h^{\mathrm{T}} \Delta_3 d_{it}^{\mathrm{T}} + \beta_h^{\mathrm{HH}} \Delta_3 d_{it}^{\mathrm{HH}} + \varepsilon_{i,t+h}, \qquad h = 0, \ldots, 5. \tag{6}$$

- **Dependent variable.** Three-year GDP growth from $t - 3 + h$ to $t + h$.
- **Interpretation.** This is the table version of the boom-bust relation: it maps a three-year credit expansion into contemporaneous and future GDP growth windows.
- **Key coefficients.** Non-tradable credit is contemporaneously positive for growth but becomes strongly negative at horizons $h = 3$ to 5. Household credit becomes even more negative. Tradable credit is positive but generally imprecise.
- **Economic reading.** All sectors can look good during the boom. The distinction only appears once one follows the boom forward into the medium run.

3.6 Predicting systemic banking crises

To connect sectoral credit growth to financial fragility, the paper estimates

$$\mathrm{Crisis}_{i,t+1 \text{ to } t+h} = \alpha_i^h + \sum_{k \in K} \beta_k^h \Delta_3 d_{it}^k + \varepsilon_{i,t+h}. \tag{7}$$

- **Dependent variable.** An indicator that equals one if a systemic banking crisis starts between $t + 1$ and $t + h$.
- **Interpretation.** This is a predictive linear-probability model for cumulative crisis risk.
- **Expected signs.** If bad booms are those concentrated in fragile sectors, then non-tradable and household credit should have positive coefficients, while tradable credit should have weak or negative coefficients.
- **Result.** Exactly that pattern appears. A two-standard-deviation rise in non-tradable credit raises the next-year crisis probability to about 5%, versus an unconditional crisis probability of 3%.
- **Model performance.** The in-sample AUC is around 0.73 in the broad sector specification and 0.75 in the more disaggregated one.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The paper is not built around a single external instrument. Its core evidence is predictive, reduced-form, and panel based.
- The main identification comes from comparing how differences in sectoral credit growth within a country predict later macro outcomes, after absorbing country fixed effects and, in some specifications, rich lag structures or country-year effects.
- For the boom-reallocation equation, adding country-year fixed effects gives a difference-in-differences flavor: within the same country-year, sectors with higher non-tradability, more small firms, or higher mortgage reliance grow faster during booms.

4.2 Why the results are nevertheless fairly persuasive

- **Many robustness checks line up.** The sectoral-growth results survive macro controls, excluding the 2008 crisis, separate advanced- and emerging-market samples, year fixed effects, growth-trend controls, and controls for sectoral value added.
- **Bond-market concern.** Because the sectoral credit data are mostly bank based, one may worry the results miss bond finance. The paper shows that adding information on bond issuance does not overturn the core patterns.
- **Credit-risk concern.** Another concern is that sectoral credit merely proxies for lower-quality issuance. The paper adds Greenwood-Hanson-style risk measures (ISS and HYS proxies) and finds little change in the main coefficients.
- **Mechanism triangulation.** The same sectoral pattern appears in boom composition, crisis prediction, sectoral NPL losses, house-price reversals, real appreciation, and productivity dynamics. That consistency helps credibility even without a single causal shock.

4.3 What the paper does *not* claim

- The paper explicitly says it cannot causally identify the exact mechanism in such a broad macro panel.
- It is better read as a disciplined empirical sorting device among theories: the data look much more consistent with easy-credit/collateral-feedback models for non-tradables than with a view that all credit expansions are equally productive.

5 Tables and figures

5.1 Table 1: Comparison with existing credit datasets

Read:

- This is the paper's data-contribution table. The authors show that their sectoral database dominates existing sources in country coverage, time span, and within-country sector detail.
- The headline number is 89,019 country-sector-year observations, versus 4,103 in Jorda et al. (2016a).
- For total credit, the database also extends long-run coverage to 189 countries, which matters because the paper wants both deep history and broad geography.

TABLE 1
Comparison with existing data sources on private credit

Dataset	Start	Freq.	Countries	Country-year obs.	Sectors	Country-sector-year obs.
Panel A: Sectoral credit data						
Müller–Verner	1940	Y	117	5,436	Mean = 16	89,019
Jordà *et al.* (2016a)	1870	Y	18	1,764	3	4,103
IMF GDD	1950	Y	83	1,871	2	3,703
BIS	1940	Q	43	1,220	2	2,417
Panel B: Total credit data						
Müller–Verner	1910	Y	189	10,272	–	10,272
IMF IFS	1948	Y/Q/M	182	8,458	–	8,458
World Bank GFDD	1960	Y	187	7,745	–	7,745
IMF GDD	1950	Y	159	6,802	–	6,802
Monnet and Puy (2019)	1940	Q	46	2,936	–	2,936
BIS	1940	Q	43	2,020	–	2,020
Jordà *et al.* (2016a)	1870	Y	18	1,816	–	1,816

Notes: Panel A compares data that differentiate between different sectors of the economy (*e.g.* household versus firm credit). Panel B compares different sources of data on total credit to the private sector. WB GFDD stands for the World Bank's Global Financial Development Database (Cihák *et al.*, 2013). BIS refers to the credit to the non-financial sector statistics described in Dembiermont *et al.* (2013). IMF IFS and GDD refer to the International Monetary Fund's International Financial Statistics and Global Debt Database (Mbaye *et al.*, 2018), respectively. The data in Monnet and Puy (2019) are from historical paper editions of the IMF IFS. *Country-year obs.* refers to the number of country-year observations covered by the datasets. *Sectors* refers to the number of covered sectors; the mean refers to the average number of sectors in a country-year panel. *Country-sector-year obs.* refers to country-sector-year observations. We count observations until 2014.

TABLE 2
Descriptive statistics

	N	Mean	Std. dev.	10th	90th			
Panel A: Summary statistics								
Real GDP growth ($t - -3, t$)	1,890	15.71	10.35	3.89	28.68			
$\Delta_3 d_{it}^k$								
Non-tradables	1,890	0.83	3.83	−2.92	5.16			
Tradables	1,890	0.03	2.26	−2.55	2.57			
Household	1,890	2.12	4.18	−1.63	7.58			
Agriculture	1,890	0.02	0.73	−0.66	0.66			
Manuf. and mining	1,890	0.01	1.87	−2.14	2.08			
Construction and RE	1,890	0.54	2.20	−1.32	3.01			
Trade, accomodation, food	1,890	0.19	1.73	−1.58	2.03			
Transport, comm.	1,890	0.11	0.75	−0.55	0.84			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Correlation matrix								
$\Delta_3 d_{it}^k$								
(1) Non-tradables	1							
(2) Tradables	0.46	1						
(3) Household	0.45	0.15	1					
(4) Agriculture	0.21	0.64	0.15	1				
(5) Manuf. and mining	0.47	0.88	0.11	0.25	1			
(6) Construction and RE	0.81	0.29	0.45	0.13	0.30	1		
(7) Trade, accom., food	0.79	0.44	0.28	0.22	0.44	0.37	1	
(8) Transport, comm.	0.55	0.29	0.22	0.084	0.32	0.29	0.33	1

Notes: Panel A shows summary statistics for the main estimation sample. Panel B plots Pearson correlation coefficients for three-year changes in the credit-to-GDP ratio $\Delta_3 d_{it}^k$ for all sectors k used in the analysis.

5.2 Table 2: Summary statistics and sector correlations

Read:

- Mean three-year GDP growth in the main sample is 15.71 log points. Mean three-year credit growth is much larger for households (2.12 percentage points of GDP) and non-tradables (0.83) than for tradables (0.03).
- Within the non-tradable block, the strongest correlations are between the aggregate non-tradable measure and construction/real estate (0.81) and trade/accommodation/food (0.79).
- This already hints that the bad-boom margin is closely tied to real-estate-related and local-demand sectors.

5.3 Figure 1: Long-run rise in private credit and the household-credit boom

Read:

- Panel (a) shows the familiar "hockey stick" for private credit-to-GDP in advanced economies, with a much more muted rise in emerging economies.
- Panel (b) shows that most of the post-1980 rise in aggregate credit comes from households, not firms.
- The figure matters because it clarifies that the global credit boom is heavily a household-credit story on one margin, but the paper's new contribution is to show that the corporate side also contains important heterogeneity.

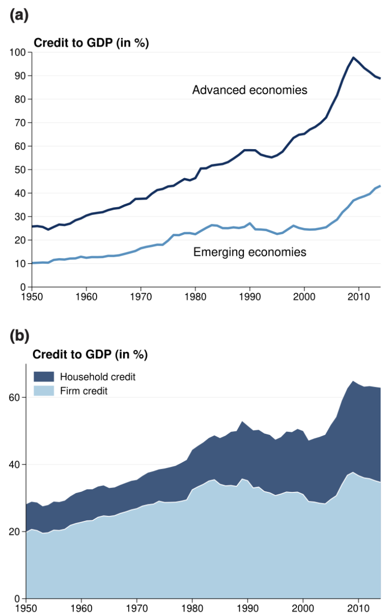

FIGURE 1
Private credit-to-GDP (in %) (a) by country group and (b) by sector, 1950–2014

Notes: Panel (a) shows the unweighted cross-country average of the ratio of total private credit-to-GDP. The average is estimated on the full sample of 58 advanced and 127 emerging economies over the period 1950–2014. Advanced economies refer to the World Bank's 2019 classification of "high-income countries," and emerging economies refers to all others. Panel (b) plots the unweighted cross-country average of sectoral credit-to-GDP. The average is estimated on the full sample of 54 advanced and 76 emerging economies, 1950–2014.

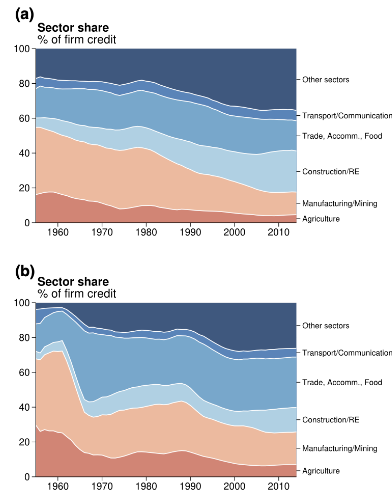

FIGURE 2
Sector shares in corporate credit: (a) advanced economies and (b) emerging economies

Notes: This figure plots the average ratio of individual sectors in total corporate credit separately for advanced and emerging economies. The plots are based on a sample of 46 advanced and 54 emerging economies. "Other sectors" is the residual of total firm credit and the sectors we use in our main analysis. This residual mainly comprises other (largely non-tradable) service sectors. Countries differ significantly in the detail of credit data reported for service sectors. To maximize the number of countries for this exercise, we group these together into "other sectors".

5.4 Figure 2: Long-run sector shares in corporate credit

Read:

- This figure shows a structural shift away from agriculture and manufacturing toward construction, real estate, and other services.
- In advanced economies, construction and real estate rise from negligible shares in the 1950s to about 24% today; other services also expand strongly.
- In emerging economies, agriculture plus industry fall from more than 73% of corporate credit in 1960 to roughly 26% today, while construction and real estate rise from around 5% to 14%.
- The point is that "business credit" is no longer synonymous with manufacturing finance.

5.5 Table 3: Why non-tradables look fragile and tradables look productive

Read:

- Non-tradable firms are smaller (small-firm share 0.90 versus 0.79), which is consistent with tighter financing constraints.
- They rely much more on real-estate collateral (0.36 versus 0.19), which links them more tightly to collateral cycles.
- Productivity growth is meaningfully lower in non-tradables: labour productivity 2.65 versus 5.03, and TFP 0.51 versus 2.02.
- This table is the micro-foundation for the paper's mechanism story: easy credit should disproportionately move non-tradables, and lending there should be more fragile and less growth enhancing.

TABLE 3

Comparing non-tradable and tradable sector characteristics

	Tradable/Non-tradable			Key industries		
	T	NT	NT–T	Manuf.	Constr./RE	Food, accomm.
Small firm share	0.79	0.90	0.12	0.78	0.91	0.86
Mortgage share	0.19	0.36	0.17	0.18	0.67	0.56
Labour productivity growth	5.03	2.65	−2.38	4.82	2.52	2.74
Total factor productivity growth	2.02	0.51	−1.51	2.19	−0.20	1.07

Notes: This table compares sectoral characteristics of non-tradable and tradable industries. *Small firm share* is defined as the share of businesses with less than 10 employees, based on the OECD Structural and Demographic Business Statistics. *Mortgage share* is the share of loans secured on real estate relative to all outstanding loans based on data from five countries: Denmark, Latvia, Switzerland, Taiwan, and the U.S. For Denmark, we use the ratio of lending by mortgage banks in each sector relative to total lending by mortgage and commercial banks, using data for 2014–20 from Danmarks Nationalbank. For Latvia, we use the share of loans secured by mortgages using data for 2006–12 from the Financial and Capital Market Commission. For Switzerland, we use the share of mortgage lending in each sector using data for 1997–2020 from the Swiss National Bank. For Taiwan, we compute the share of lending for real estate purposes in each sector using data for 1997–2015 from the Central Bank of the Republic of China (Taiwan). For the U.S., we construct the weighted average ratio of mortgages and other secured debt (`dm`) to total long-term debt (`dltt`) using Compustat. *Labour productivity growth* is defined as the average yearly percentage growth in value added per engaged person in 2005 PPP USD, calculated based on data from EU KLEMS, WIOD, and OECD STAN, as well as data on sectoral relative prices from GGDC. The estimates are based on data from 39 countries. *Total factor productivity growth* is from EU KLEMS and is based on data from 18 countries.

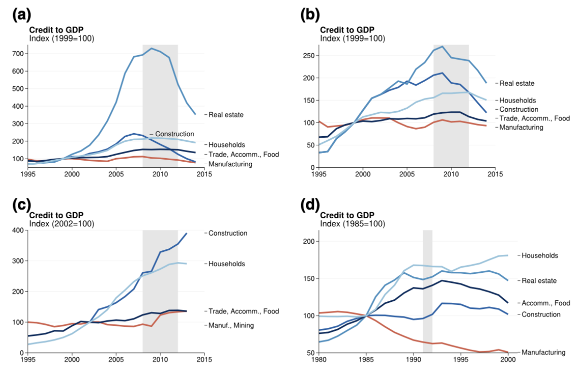

FIGURE 3

Case studies: the Eurozone and Japanese Crises. (a) Eurozone Crisis: Spain. (b) Eurozone Crisis: Portugal. (c) Eurozone Crisis: Greece. (d) Japanese Financial Crisis

Notes: Panels (a)–(c) plot the ratio of sectoral credit-to-GDP for construction (ISIC Rev. 4, section F), real estate (L), trade/accommodation/food (G + I), manufacturing (C), and households in Spain, Portugal, and Greece around the time of the Global Financial Crisis and Eurozone crisis. Values for Spain and Portugal are indexed to 100 in 1999 (the year the euro was introduced), while Greece is indexed to 100 in 2002, as construction credit data only start in that year. Panel (d) plots the ratio of sectoral credit-to-GDP for construction (ISIC Rev. 4, section F), real estate (L), trade/accommodation/food (G + I), manufacturing (C), and households around the Japanese banking crisis of the early 1990s. The areas shaded in grey mark years the countries were in a systemic banking crisis as defined by Laeven and Valencia (2018).

5.6 Figure 3: Case studies — Eurozone periphery and Japan

Read:

- Spain, Portugal, Greece, and late-1980s Japan all display the same basic pattern: rapid growth in real-estate-related and household credit, stagnation or decline in manufacturing credit, and then crisis.
- The Eurozone panels show especially dramatic surges in real-estate credit; the Japan panel shows a strong boom in real estate, accommodation/food, and households with weak manufacturing lending.
- These case studies visually motivate the paper's broader claim that bad booms are sectorally concentrated.

5.7 Figure 4 and Table 4: Credit booms reallocate credit toward non-tradables

TABLE 4

Credit booms and credit reallocation

	Dependent var.: $100 \times \Delta_3 \ln(D_{i,s,t}/\mathrm{GDP}_{i,t})$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\mathrm{Boom}_{i,t}$	23.0***	18.8***		18.7***		19.9***	
	(1.73)	(2.57)		(1.95)		(2.42)	
$\mathrm{Boom}_{i,t} \times \mathrm{Non\text{-}tradable}_s$		7.08**	7.04***				
		(3.15)	(2.57)				
$\mathrm{Boom}_{i,t} \times \mathrm{HighSmallFirmShare}_s$				7.12***	6.96***		
				(2.15)	(1.83)		
$\mathrm{Boom}_{i,t} \times \mathrm{HighMortShare}_s$						5.10**	5.33**
						(2.32)	(2.01)
Country FE	✓	✓		✓		✓	
Industry FE	✓	✓		✓		✓	
Country×Year			✓		✓		✓
Country×Industry FE			✓		✓		✓
Observations	10,529	10,529	10,526	10,529	10,526	10,529	10,526
# Countries	76	76	76	76	76	76	76
R^2	0.11	0.11	0.59	0.11	0.59	0.11	0.59

Notes: This table presents estimates of equation (1) in a country-sector-year panel, where the dependent variable is the log change (times 100) in sectoral credit-to-GDP over the previous three years. There are five sectors: agriculture; manufacturing and mining; construction and real estate; wholesale and retail trade, accommodation, and food services; and transport and communication. Non-tradable industries are: construction and real estate; wholesale and retail trade, accommodation, and food services; and transport and communication. Non-tradable industries are classified based on the inverse of the exports to value-added ratio in the U.S. Industries with a high small firm share are: agriculture; construction and real estate; and transport and communication. High mortgage share industries are: agriculture; construction and real estate; and wholesale and retail trade, accommodation, and food services. See Table 3 for details on the industry characteristics. Driscoll and Kraay (1998) standard errors with six lags in parentheses. *, **, and *** denote significance at the 10, 5, and 1% level.

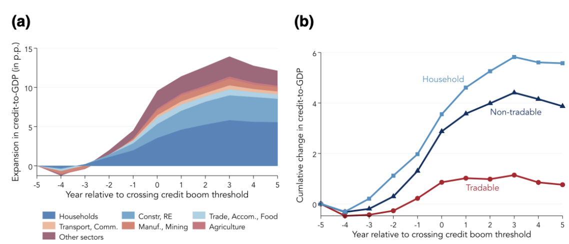

FIGURE 4

The allocation of credit during credit booms. (a) Sectoral breakdown of aggregate credit expansion. (b) Non-tradable, tradable, and household sectors

Notes: This figure plots an event study of the cumulative change in private credit-to-GDP around credit booms, broken down by sectors. Panel (a) presents the disaggregated industries, and panel (b) shows the non-tradable, tradable, and household sector aggregates. The credit boom events are defined as periods of large deviations from a (Hamilton, 2018) filter with a horizon of four years. The change in credit-to-GDP in each sector is demeaned at the country level to abstract from longer-term trends in credit over time within countries. "Other sectors" is a residual category that includes services not included in the remaining industries.

Read:

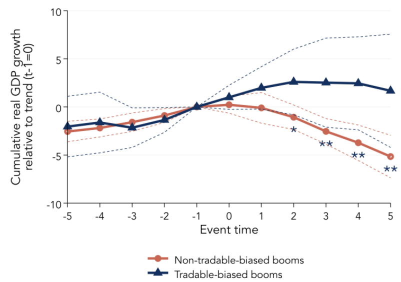

FIGURE 5

Output dynamics around tradable and non-tradable-biased credit booms

Notes: This figure plots results from estimating equation (2). Time zero is defined as the first year in which the credit boom is identified. Tradable-biased (non-tradable-biased) credit booms are defined as booms in which the share of tradable-sector credit (non-tradable and household sector credit) rises from time $t = -5$ to $t = 0$. The union of $\mathbf{Boom}_{it}^{\mathrm{T}}$ and $\mathbf{Boom}_{it}^{\mathrm{NT}}$ thus comprises all identified credit booms. Dashed lines represent 90% confidence intervals based on Driscoll and Kraay (1998) standard errors with lag length $ceiling(1.5(3+h))$. *, **, and *** indicate that the difference between the estimates, $\hat{\beta}_{\mathrm{T}}^{h} - \hat{\beta}_{\mathrm{NT}}^{h}$, is statistically significant at the 10, 5, and 1% level, respectively.

- Figure 4's event study shows that households are the largest contributor to boom-time credit growth, followed by construction/real estate and trade/accommodation/food.
- The non-tradable plus household sectors account for roughly 70% of total lending growth during major booms.
- Table 4 formalizes this: during credit booms, three-year sectoral credit growth is about 7.1% higher in non-tradables, 7.1% higher in high-small-firm sectors, and 5.1% higher in high-mortgage-share sectors.
- The country-year fixed-effect versions are especially important because they show that, within the same macro boom, credit tilts toward fragile sectors.

5.8 Figure 5: Tradable-biased versus non-tradable-biased booms

Read:

- Both boom types are expansionary on the way up, but only non-tradable-biased booms experience a strong reversal after the peak.
- Five years after boom start, GDP is about 5% lower following a non-tradable-biased boom than following a tradable-biased boom.
- This figure is the paper's sharpest "good booms versus bad booms" exhibit.

5.9 Figure 6 and Table 5: GDP dynamics after sectoral credit expansions

Read:

- Figure 6's local projections show that non-tradable credit predicts lower future GDP, while tradable credit predicts stable or higher future GDP.
- In Table 5, non-tradable credit is contemporaneously growth positive (0.46 in Panel A, column 1) but becomes strongly negative in the medium run: -0.38, -0.47, and -0.43 over the $(t, t+3)$, $(t+1, t+4)$, and $(t+2, t+5)$ windows.
- Household credit becomes even more negative once future windows are considered: -0.39, -0.53, and -0.55 in Panel B.

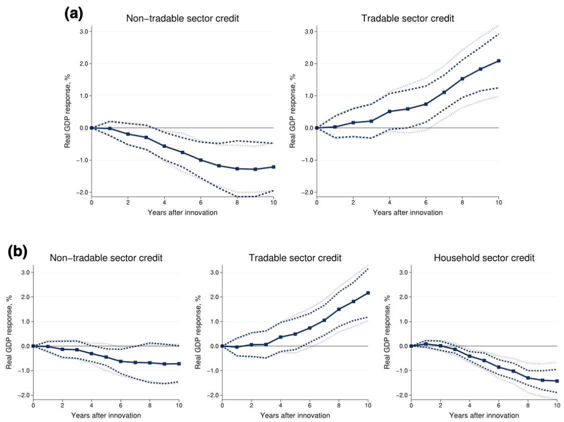

FIGURE 6

Output dynamics after credit expansions in tradable, non-tradable, and household sectors. (a) Non-tradable and tradable sector credit. (b) Non-tradable, tradable, and household sector credit

Notes: This figure presents local projection impulse responses of real GDP following innovations in tradable sector credit, non-tradable sector credit, and household credit (all measured relative to GDP). The impulse responses are based on estimation of equation (3). Panel (a) includes non-tradable and tradable firm credit, while panel (b) presents results from the same specification that also includes household credit. Dashed lines represent 95% confidence intervals computed using Driscoll and Kraay (1998) standard errors, and dotted lines represent 95% confidence intervals from standard errors two-way clustered on country and year.

TABLE 5

Sectoral credit expansion and GDP growth

	Dependent var.: GDP growth over…					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Non-tradable and tradable sector credit						
$\Delta_3 d_{it}^k$	$(t-3, t)$	$(t-2, t+1)$	$(t-1, t+2)$	$(t, t+3)$	$(t+1, t+4)$	$(t+2, t+5)$
Tradables	0.087	0.11	0.19	0.30	0.38	0.39
✓	(0.15)	(0.18)	(0.18)	(0.19)	(0.24)	(0.26)
Non-tradables	0.46***	0.15	−0.18*	−0.38***	−0.47***	−0.43***
✓	(0.088)	(0.11)	(0.10)	(0.11)	(0.14)	(0.16)
Observations	1,890	1,820	1,748	1,677	1,605	1,533
# Countries	75	75	75	75	75	75
R^2	0.05	0.01	0.01	0.02	0.03	0.03
Panel B: Including household credit						
$\Delta_3 d_{it}^k$	$(t-3, t)$	$(t-2, t+1)$	$(t-1, t+2)$	$(t, t+3)$	$(t+1, t+4)$	$(t+2, t+5)$
Tradables	0.086	0.095	0.16	0.26	0.33	0.34
✓	(0.14)	(0.17)	(0.17)	(0.17)	(0.20)	(0.23)
Non-tradables	0.47***	0.21*	−0.045	−0.19**	−0.23***	−0.19**
✓	(0.075)	(0.11)	(0.100)	(0.079)	(0.069)	(0.091)
Households	−0.0070	−0.11	−0.25***	−0.39***	−0.53***	−0.55***
✓	(0.090)	(0.088)	(0.075)	(0.071)	(0.10)	(0.13)
Observations	1,890	1,820	1,748	1,677	1,605	1,533
# Countries	75	75	75	75	75	75
R^2	0.05	0.01	0.02	0.05	0.08	0.08

Notes: This table presents the results from estimating equation (4). The dependent variable in column h is the change in log real GDP (times 100) from year $t-3+h$ to $t+h$. The right-hand side variables, $\Delta_3 d_{it}^k$, are the changes in the credit/GDP ratio (in percentage points) for sector k from $t-3$ to t. Driscoll and Kraay (1998) standard errors in parentheses with lag length $ceiling(1.5(3+h))$. *, **, and *** denote significance at the 10, 5, and 1% level.

- Tradable credit, by contrast, is positive at all horizons, though imprecisely estimated. This is exactly the pattern the paper wants: bad medium-run predictability is concentrated in non-tradables and households.

5.10 Figure 7 and Table 6: The crisis link

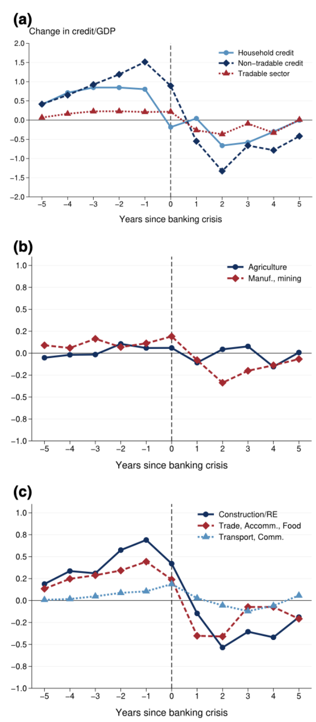

FIGURE 7

Credit dynamics around systemic banking crises. (a) Tradable versus non-tradable sector. (b) Tradable sector industries. (c) Non-tradable sector industries

Notes: This figure plots average annual percentage point changes in sectoral credit-to-GDP ratios around 59 systemic banking crises in 90 countries between 1951 and 2009. The horizontal axis represents the number of years before and after a crisis. Crisis dates are from Baron *et al.* (2021), supplemented with dates from Laeven and Valencia (2018) for countries where they report no data.

TABLE 6

Sectoral credit expansions and financial crises

	Dependent variable: Crisis within…			
Panel A: Non-tradable, tradable, and household sector credit				
$\Delta_3 d_{it}^k$	1 year	2 years	3 years	4 years
Tradables	−0.004*	−0.007**	−0.007*	−0.004
	(0.002)	(0.004)	(0.003)	(0.004)
Non-tradables	0.006***	0.010***	0.011***	0.008*
	(0.002)	(0.002)	(0.003)	(0.004)
Household	0.004**	0.008**	0.010**	0.012***
	(0.002)	(0.003)	(0.004)	(0.004)
Observations	1,557	1,557	1,557	1,557
# Countries	72	72	72	72
# Crises	47	47	47	47
Mean crisis prob.	0.03	0.06	0.09	0.12
ΔProb. if 2 SD higher $\Delta_3 d_{it}^{NT}$	0.05	0.08	0.08	0.06
AUC	0.73	0.70	0.69	0.67
SE of AUC	0.04	0.03	0.03	0.02
Panel B: Individual corporate sectors				
$\Delta_3 d_{it}^k$	1 year	2 years	3 years	4 years
Agriculture	−0.007	−0.008	−0.010	−0.010
	(0.004)	(0.009)	(0.016)	(0.017)
Manuf. and mining	−0.003	−0.008	−0.007	−0.003
	(0.003)	(0.005)	(0.005)	(0.006)
Construction and RE	0.008***	0.012***	0.011***	0.008
	(0.003)	(0.005)	(0.004)	(0.005)
Trade, accomodation, food	0.009***	0.020***	0.026***	0.028***
	(0.003)	(0.005)	(0.008)	(0.008)
Transport, comm.	−0.008	−0.018**	−0.032***	−0.044**
	(0.006)	(0.007)	(0.010)	(0.019)
Household	0.004**	0.007**	0.010***	0.012***
	(0.002)	(0.003)	(0.003)	(0.003)
Observations	1,557	1,557	1,557	1,557
# Countries	72	72	72	72
# Crises	47	47	47	47
Mean crisis prob.	0.03	0.06	0.09	0.12
AUC	0.75	0.74	0.72	0.71
SE of AUC	0.04	0.03	0.02	0.02

Notes: This table presents the results from estimating equation (5). In Panel A, we differentiate between credit to the tradable, non-tradable, and household sectors. In Panel B, we use individual corporate sectors. Crisis dates are from Baron *et al.* (2021), supplemented with dates from Laeven and Valencia (2018) for countries not covered by Baron *et al.* (2021). Driscoll and Kraay (1998) standard errors with lag length $ceiling(1.5(3+h))$ are in parentheses. *, **, and *** denote significance at the 10, 5, and 1% level.

Read:

- Figure 7 shows that banking crises are preceded by fast growth in non-tradable and household credit, especially in construction/real estate and trade/accommodation/food.
- Table 6 shows that non-tradable and household credit expansions significantly raise future crisis probabilities, while tradable credit slightly lowers them.
- A two-standard-deviation rise in non-tradable credit raises next-year crisis probability from 3% to 5%. The AUC of 0.73 to 0.75 means sectoral credit has substantial crisis-classification power.
- Within the corporate sector, construction/real-estate credit and trade/accommodation/food credit are the main crisis-predictive components.

5.11 Figure 8: Sectoral loan losses after crises

Read:

- This figure is the paper's strongest direct evidence on banking-sector losses.
- At peak NPL years after crises, NPL ratios are about 50% higher in the non-tradable sector than in the tradable sector.
- Non-tradables plus households account for nearly three-quarters of total NPLs; non-tradables alone account for roughly half.
- This links pre-crisis sectoral credit reallocation directly to post-crisis bank distress.

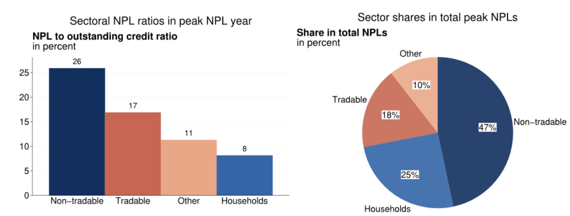

FIGURE 8

Financial crises and sectoral loan losses: evidence from ten banking crises

Notes: This figure documents sectoral differences in loan losses and the composition of NPLs following ten systemic banking crises. The included crisis episodes (based on data availability) are Mexico (1994), Thailand (1997), Indonesia (1997), Turkey (2000), Argentina (2001), Italy (2008), Latvia (2008), Croatia (2008), Spain (2008), and Portugal (2008). Note that Laeven and Valencia (2018) do not classify Croatia as experiencing a crisis in 2008, but Croatia did experience a long-lasting recession following a period of rapid capital inflows and growth in corporate debt. The left panel shows the median ratio of NPLs to outstanding loans separately for the non-tradable, tradable, and household sectors in the peak NPL year. The right panel plots the median share of the individual sectors in total NPLs in the year where the total NPL ratio reached its peak (within ten years after each crisis).

TABLE 7

Sectoral credit expansions and house price growth

	Dependent var.: Real house price growth over…					
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_3 d_{it}^k$	$(t-3,t)$	$(t-2,t+1)$	$(t-1,t+2)$	$(t,t+3)$	$(t+1,t+4)$	$(t+2,t+5)$
Tradables	0.63	0.88*	1.14**	1.06**	1.13***	0.85***
	(0.47)	(0.51)	(0.54)	(0.50)	(0.41)	(0.26)
Non-tradables	0.95***	0.16	−0.68*	−1.16***	−1.33***	−1.04***
	(0.26)	(0.34)	(0.35)	(0.32)	(0.20)	(0.19)
Households	0.49	0.16	−0.20	−0.64***	−0.95***	−0.90***
	(0.33)	(0.34)	(0.21)	(0.15)	(0.15)	(0.20)
Observations	895	881	864	847	829	810
# Countries	42	42	42	42	42	42
R^2	0.11	0.02	0.03	0.09	0.14	0.10

Notes: This table presents the results from estimating equation (4) with $\Delta_3 \ln(\text{HPI})_{it+h}$ (the three-year change in log real house prices) as the dependent variable. All columns include country fixed effects. Driscoll and Kraay (1998) standard errors in parentheses with lag length $ceiling(1.5(3+h))$. *, **, and *** denote significance at the 10, 5, and 1% level.

5.12 Table 7: House-price booms and busts

Read:

- All sectoral credit expansions co-move positively with contemporaneous house-price growth, but the strongest contemporaneous link is for non-tradables.
- The medium-run reversal is again concentrated in non-tradables and households. A one-standard-deviation rise in non-tradable credit predicts about 5.1 percentage points lower house-price growth from t to $t+3$; the analogous number for household credit is about 3.0 points.
- Tradable credit, in contrast, predicts stronger future house prices rather than a bust.

TABLE 8

Sectoral credit expansions, sectoral real activity, and the real exchange rate

	(1)	(2)	(3)
$\Delta_3 d_{it}^k$	$\Delta_3 \ln\left(\frac{Y^{\mathrm{NT}}}{Y^{\mathrm{T}}}\right)$	$\Delta_3 \ln\left(\frac{E^{\mathrm{NT}}}{E^{\mathrm{T}}}\right)$	Δ_3 ln (RER)
Tradables	0.29	−0.30	−0.32
	(0.23)	(0.19)	(0.29)
Non-tradables	0.70***	0.43***	0.43**
	(0.16)	(0.076)	(0.20)
Households	0.41***	0.27***	0.31**
	(0.10)	(0.058)	(0.12)
Observations	1,638	846	1,793
# Countries	69	36	75
R^2	0.09	0.14	0.03

Notes: This table presents regressions of changes in various macroeconomic outcomes from $t-3$ to t on the expansion in tradable, non-tradable, and household credit-to-GDP over the same period. The outcome variables are the log of the non-tradable to tradable value-added ratio (column (1)), the log of the non-tradable to tradable employment ratio (column (2)), and the log of the real effective exchange rate (column (3)). The real effective exchange rate is defined such that an increase signifies real appreciation. All columns include country fixed effects. Driscoll and Kraay (1998) standard errors in parentheses with lag length of 6. *, **, and *** denote significance at the 10, 5, and 1% level.

TABLE 9

Sectoral credit expansions and labour productivity growth

	Dependent variable: Labour productivity growth over…					
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_3 d_{it}^k$	$(t-3, t)$	$(t-2, t+1)$	$(t-1, t+2)$	$(t, t+3)$	$(t+1, t+4)$	$(t+2, t+5)$
Tradables	0.186*	0.173**	0.208**	0.219*	0.199	0.169
	(0.093)	(0.081)	(0.086)	(0.113)	(0.150)	(0.179)
Non-tradables	0.066	−0.072	−0.168**	−0.147**	−0.080	−0.009
	(0.142)	(0.125)	(0.082)	(0.067)	(0.054)	(0.059)
Households	−0.147**	−0.183***	−0.226***	−0.261***	−0.312***	−0.308***
	(0.061)	(0.061)	(0.057)	(0.067)	(0.076)	(0.068)
Observations	1,451	1,451	1,451	1,451	1,451	1,451
# Countries	69	69	69	69	69	69
R^2	0.01	0.01	0.03	0.03	0.03	0.03

Notes: This table presents the results from estimating equation (4) with the three-year change in the log of labour productivity as the dependent variable. Driscoll and Kraay (1998) standard errors in parentheses with lag length $ceiling(1.5(3+h))$. *, **, and *** denote significance at the 10, 5, and 1% level.

5.13 Table 8 and Table 9: Sectoral imbalances, real appreciation, and productivity

Read:

- Table 8 shows that non-tradable and household credit expansions coincide with a reallocation of value added and employment toward non-tradables and with real exchange rate appreciation.
- Non-tradable credit has coefficients of 0.70 for the non-tradable/tradable value-added ratio, 0.43 for the employment ratio, and 0.43 for the real exchange rate.
- Table 9 shows the growth side of the distinction: tradable credit predicts higher future labour-productivity growth, while non-tradable and especially household credit predict lower future productivity growth.
- In Table 9, tradable credit has positive coefficients of about 0.17 to 0.22 across the main windows, whereas non-tradable credit turns negative and household credit is strongly negative throughout.

6 Short summary

- **Conclusion.** The paper shows that the macroeconomic consequences of credit booms depend crucially on sectoral allocation. Credit to non-tradables and households predicts boom-bust dynamics and crises; credit to tradables does not.
- **Intuition.** Easy credit disproportionately flows to smaller, collateral-dependent, domestic-demand-sensitive sectors. Those sectors boom when financing is easy and crack when financing tightens or collateral prices fall.
- **Empirical lesson.** Treating all firm credit as one aggregate hides the key margin of heterogeneity. Distinguishing among sectors is essential for understanding both financial fragility and the growth consequences of credit.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The allocation of credit across sectors is a central predictor of whether a credit boom is benign or fragile.

- **Bad booms.** Credit booms tilted toward non-tradable firms and households are followed by GDP slowdowns, house-price busts, banking crises, and concentrated bank losses.
- **Good booms.** Credit expansion toward tradable firms is associated with stable or stronger future output and higher productivity growth, without higher crisis risk.
- **Mechanism evidence.** Non-tradable firms are smaller, more dependent on real-estate collateral, lower-productivity, and more likely to default in crises. The mechanism evidence consistently points toward easy-credit and collateral-feedback channels.
- **Methodological contribution.** The paper's new sectoral-credit database makes it possible to go beyond the household-versus-firm split and distinguish among types of corporate credit.
- **Policy implication.** If taken at face value, the evidence supports a more sectoral view of macroprudential policy: risks are not only about total credit growth, but also about *where* credit is flowing.

7.2 Economic intuition

- Non-tradable firms are exactly the firms that benefit the most when lenders become optimistic or collateral values rise: they are smaller, more opaque, more domestically oriented, and more tied to real estate.
- That same dependence makes them fragile in reverse. When credit supply tightens, asset prices fall, or domestic demand weakens, these firms contract sharply and default more.
- Tradable-sector borrowing looks different. It is more naturally tied to productivity, external demand, and sectors with higher long-run efficiency growth. Credit there behaves more like investment finance than speculative leverage.
- The macroeconomy therefore inherits the sectoral composition of credit. The same aggregate credit boom can be good or bad depending on whether it finances productivity-enhancing tradables or fragile non-tradables and households.

7.3 Takeaway

This paper's main insight is simple and powerful: not all credit is created equal. Aggregate credit growth is too coarse a statistic for understanding macro-financial risk. Once credit is decomposed by sector, a clear pattern emerges. Lending to non-tradables and households is the margin associated with boom-bust cycles and crises, while lending to tradables is the margin more closely associated with sustained growth. That is why sectoral credit allocation belongs at the center of any serious theory of credit cycles.